SECTION A [60 *marks*]

Answer all questions in this section.

1 Compound Q contains carbon, hydrogen and nitrogen. Combustion of 0.250 g of Q produces 0.344 g of water, H_2O, and 0.558 g of carbon dioxide, CO_2. Determine

(a) the empirical formula of Q,

[11 *marks*]

(b) the molar mass of Q if the molecular formula is the same as the empirical formula and

[1 *marks*]

(c) the number of hydrogen atoms present in the above sample Q.

[3 *marks*]

2 An atom X has 5 valence electrons. X reacts with fluorine gas to form XF_3 and XF_5 compounds.

(a) For each compound,

(i) draw the Lewis structure,

(ii) predict the electron pair geometry and the molecular geometry and

(iii) draw the molecular geometry and state the bond angle(s).

[13 *marks*]

(b) Predict the change in hybridization (if any) of the X atom in the following reaction:

$$XF_3 + F_2 \rightarrow XF_5$$

[2 *marks*]

3 (a) A gas mixture contains 82% (w/w) methane, CH_4, and 18% (w/w) ethane, C_2H_6. If a 15.50 g sample of the gas mixture is placed in a 15-L container at 20.0°C, calculate

(i) the total moles of gases,

(ii) the total pressure (atm) and

(iii) the partial pressure of each gas in the container.

[7 *marks*]

(b) Explain each of the following observations:

(i) Methane, CH_4 (16 g mol^{-1}) has lower boiling point than propane, C_3H_8 (44 g mol^{-1}) whereas water, H_2O (18 g mol^{-1}) has higher boiling point than hydrogen sulphide, H_2S (34 g mol^{-1}).

(ii) Carbon dioxide, CO_2, molecule and diamond are both covalently bonded. However, solid CO_2, is softer and exhibits lower melting point as compared to diamond.

[8 *marks*]

4 (a) **FIGURE 1** shows a titration curve for 20.0 mL of an unknown concentration of NaOH solution titrated against a standard solution of 1.00 M HCl.

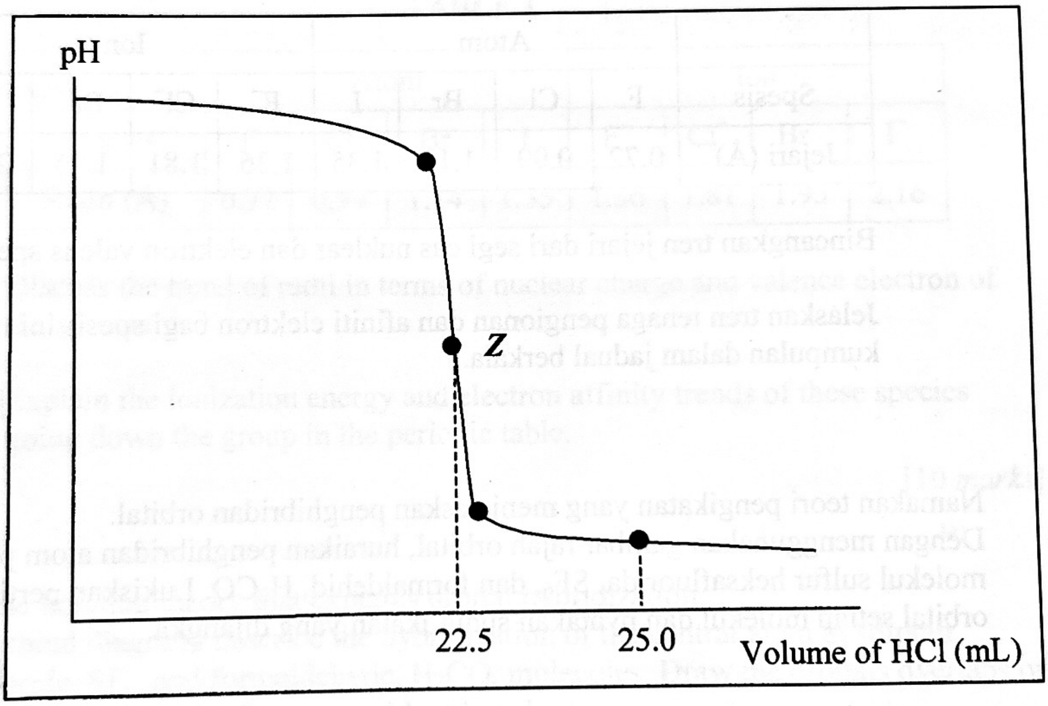

FIGURE 1

(i) Define standard solution, equivalent point and end point.

(ii) What is the pH at equivalent point, ***Z*** for this titration?

(iii) Calculate the molarity of the NaOH solution used in the titration.

(iv) Calculate the pH of the solution after the addition of 25.0 mL of HCl.

[8 *marks*]

(b) Pyridine, C_5H_5N, is a weak base. This base ionises in water to produce $C_5H_5NH^+$ and OH^-. Calculate the pH of a 1.00 M pyridine solution if the dissociation constant, K_b, of pyridine is 1.50×10^{-9}.

[7 *marks*]

SECTION B [40 *marks*]

*Answer only **two** questions in this section.*

5 (a) Calculate the first **three** wavelengths of the possible transitions of an electron in the Paschen series for a hydrogen atom. Show these wavelengths in a sketch of the line spectrum for this emission series. Explain how these transitions occurred.

[10 *marks*]

(b) The following data are given for atomic and ionic radii of halogens and halides respectively.

TABLE 1

	Atom				Ion			
Species	F	Cl	Br	I	F^-	Cl^-	Br^-	I^-
Radii (Å)	0.72	0.99	1.14	1.35	1.36	1.81	1.95	2.16

Discuss the trend of radii in terms of nuclear charge and valence electron of the species.

Explain the ionization energy and electron affinity trends of these species going down the group in the periodic table.

[10 *marks*]

6 Name the bonding theory that explains orbital hybridization.
Using orbital diagram, describe the hybridization of the central atom in sulphur hexafluoride, SF_6, and formaldehyde, H_2CO, molecules. Draw the orbitals overlaps of each molecule and state the expected bond angles.

[20 *marks*]

7 (a) Crystalline solid is classified based on the assembly of their respective basic particles. State **four** types of crystalline solid and discuss the nature of the particle assembly in each crystalline solid. Give **one** example for each type of the crystalline solid.

[10 *marks*]

(b) Consider the following reaction system;

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

This system is made up from 2.0×10^{-3} mol of each gaseous component in a 2-L closed vessel. If the equilibrium constant, K_c, for the system is 7.7×10^{7} at 600 K, prove that the system is not in equilibrium at this temperature.

When the system is cooled to 450 K and left to reach equilibrium, the final amount of O_2 detected in the vessel is 9.6×10^{-4} M. Calculate the equilibrium constant, K_c, of the system at 450 K.

[10 *marks*]

8 (a) Milk of magnesia consists of saturated solution of $Mg(OH)_2$ and gelatinous $Mg(OH)_2$. Determine the molarity of Mg^{2+} in a saturated solution of $Mg(OH)_2$ and in the saturated solution of $Mg(OH)_2$ containing 0.01 M NaOH.

[Given: K_{sp} $Mg(OH)_2 = 8.9 \times 10^{-12}$]

[6 *marks*]

(b) A buffer solution is prepared by dissolving 0.125 mol of sodium nitrite, $NaNO_2$, in 500 mL of 0.25 M nitrous acid, HNO_2. What is the pH of the solution?

The above buffer solution can resist the changes in pH when a small amount of strong acid or base is added into it. Explain how this buffer solution maintains its pH. If an amount of 0.001 mol of NaOH is added to the solution, determine the change in the pH value of the buffer solution.

[Given: K_a $HNO_2 = 5.10 \times 10^{-4}$]

[14 *marks*]

END OF QUESTION PAPER